CLINICAL · EAR

Acute, coalescent and masked mastoiditis

In acute mastoiditis the infection has broken out of the middle ear into the bone behind it. The typical patient is a small, unwell child with a red, tender swelling behind an ear that is pushed forward; the dangerous one is the patient on antibiotics whose ear has gone quiet while the bone is still being destroyed.

AT A GLANCE

- Otitis media plus a tender, swollen mastoid and a displaced ear.
- Admit, culture, IV antibiotics, myringotomy.
- Opaque mastoid isn't enough: look for lost septa or abscess.
- Operate for abscess, coalescence, failure or complication.
- Repeated antibiotics can hide it.

1 CHAPTER ONE Overview

When otitis media escapes the middle ear.

1.1 Definition

Every acute otitis media inflames the mastoid air cells to some degree, because the antrum is continuous with the middle ear; a CT in plain otitis media often shows an opaque mastoid. *Acute mastoiditis* means clinically evident infection of the mastoid process, and *coalescent mastoiditis* means the bony septa between air cells have been destroyed, turning the cells into a single cavity full of pus.

1.2 Who gets it

Half of children with acute mastoiditis are four years old or younger, and only about 30% have a history of recurrent otitis media; often it is the first severe ear infection.

1.3 Why it matters

Mastoiditis is the gateway to the other complications of otitis media: neck and facial abscesses, facial palsy, labyrinthitis and intracranial infection. Up to one in ten develop an intracranial complication, even on intravenous antibiotics.

CHAPTER TWO

2 Pathology

From inflamed mucosa to destroyed bone.

2.1 Organisms

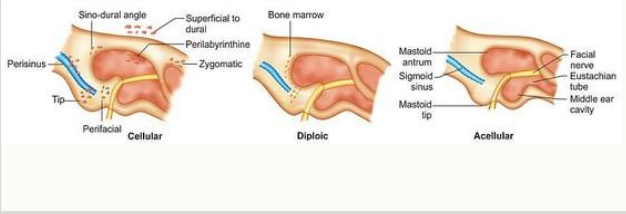
Streptococcus pneumoniae leads, with *Streptococcus pyogenes*, *Staphylococcus aureus* and *Haemophilus influenzae*. Pneumococcal serotype 19A is both resistant and virulent. Cultures are negative in many children who have already had antibiotics.

WHERE THE BOOKS DIFFER · COMMONEST ORGANISM

Dhingra	Lee
Beta-haemolytic streptococcus.	Streptococcus pneumoniae.
Lee reflects more recent series.	

2.2 How the bone is destroyed

Two processes run together. Pus under pressure in the cells causes ischaemic necrosis of the thin bony septa, and inflamed mucosa activates osteoclasts that resorb bone (*hyperaemic decalcification*). The cells coalesce into an abscess cavity, which then looks for a way out.



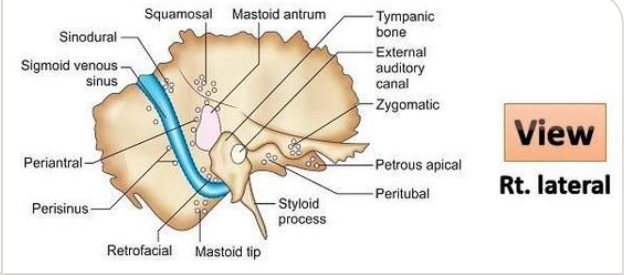


FIGURE 1 Types of mastoid (cellular, diploic, acellular) and the groups of air cells through which infection spreads.

Anatomy atlas · ENT Steps p. 59, ENT Steps p. 60

2.3 Classification

MASTOIDITIS

- ACUTE **Acute mastoiditis** otitis media with mastoid inflammation
- BONE LOSS **Coalescent** septa destroyed, pus in a single cavity
- AFTER ANTIBIOTICS **Masked (latent)** slow bone destruction, few symptoms
- LONG-STANDING **Chronic** with perforation, cholesteatoma or grommet

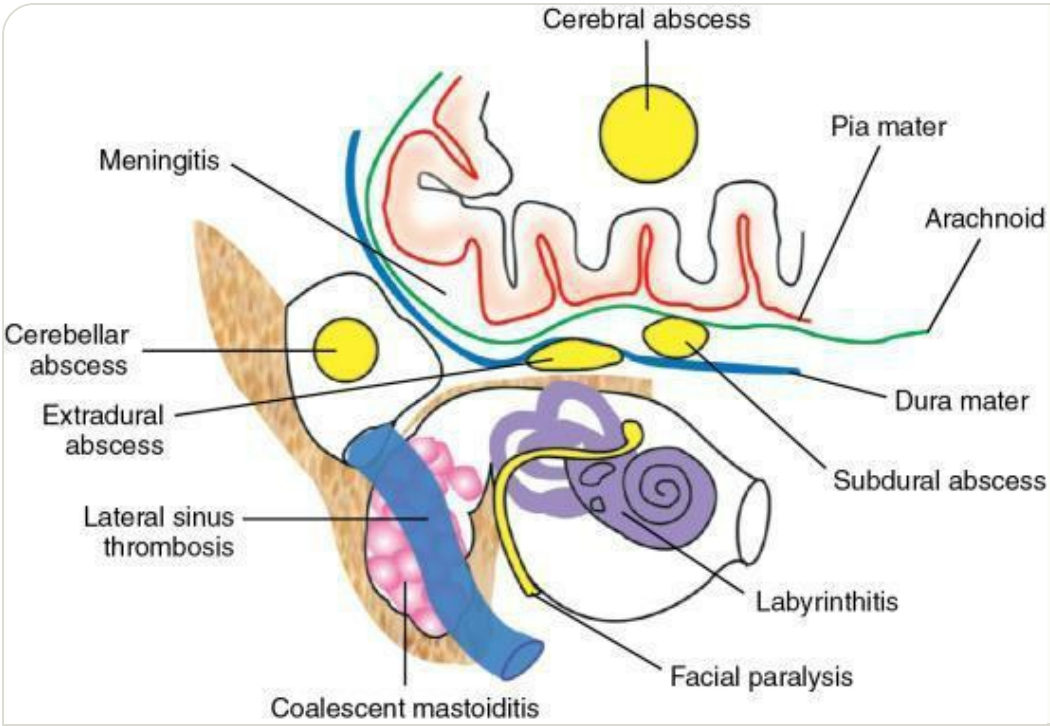


FIGURE 2 Routes by which suppurative otitis media spreads to cause its complications.
Dhingra, fig. 12.1

3 CHAPTER THREE Clinical features

The child, the ear and the bone.

3.1 Symptoms

3.1.1 Pain

Persistent pain behind the ear, often after an apparently improving otitis media.

3.1.2 Fever

Fever that persists or returns despite treatment.

3.1.3 Discharge

Ear discharge that becomes profuse and creamy, or persists beyond three weeks. Mucoid discharge must come from the middle ear, because the canal has no mucous glands.

THE CHANGE IS WHAT MATTERS

Pain, fever or discharge that continues or comes back during an acute otitis media points to mastoiditis.

3.2 Signs

3.2.1 The mastoid

Tenderness is most telling over the tip and posterior border, rather than over the suprameatal (MacEwen's) triangle, which is tender in plain otitis media too. Always compare with the healthy side. Later the area becomes red, boggy and fluctuant.

3.2.2 The pinna

A subperiosteal abscess pushes the auricle forwards and downwards, and the postauricular sulcus is lost.

3.2.3 The canal and drum

- **Sagging** of the posterosuperior canal wall, from periostitis over the antrum.
- **Reservoir sign:** the canal refills with pus straight after mopping.
- A pars tensa perforation with pulsatile discharge, sometimes with a *light-house* reflex.
- Hearing loss is conductive.



FIGURE 3 Acute mastoiditis: the pinna is pushed forward and downward by the swelling behind it.

Dhingra, fig. 12.2

3.3 Masked mastoiditis

Repeated or inadequate antibiotic courses can sterilise the middle ear while bone destruction continues. The patient has little pain and no discharge, or only a mild ache or persistent otalgia. The drum is dull and thickened with lost landmarks, and hearing is reduced. It may hide an extradural or perisinus abscess.

WHERE THE BOOKS DIFFER · SYMPTOMS OF MASKED MASTOIDITIS

Dhingra	Lee
Painless, without discharge; mild ache at most.	Persistent, chronic otalgia after antibiotics.

Either way, a patient whose ear never quite settles after antibiotics needs a CT.

4 CHAPTER FOUR Abscesses

Where the pus comes out.

4.1 Subperiosteal (postauricular)

The commonest. Pus breaks through the cribriform area into the space under the periosteum over the mastoid, displacing the pinna.

4.2 Zygomatic

Infection of the zygomatic air cells produces swelling in front of and above the ear, with oedema of the upper eyelid.

4.3 Bezold abscess

In an adult or older child with a pneumatised tip, pus breaks through the thin medial side of the tip and tracks under the sternocleidomastoid, presenting as a tender upper neck mass.

4.3.1 Differential

Upper jugular lymphadenitis, a parotid mass, an infected branchial cyst, parapharyngeal abscess and jugular vein thrombosis.

4.3.2 Treatment

Cortical mastoidectomy, finding the fistula at the tip, and drainage of the neck through a separate incision.

4.4 Meatal (Luc) abscess

Pus from the antrum tracks between the posterior canal wall and the periosteum, presenting as a swelling in the deep canal.

4.5 Citelli abscess

Pus from the tip tracks along the posterior belly of digastric to form an abscess behind the mastoid or in the digastric triangle.

4.6 Deep neck

Parapharyngeal and retropharyngeal abscesses via the tip cells or petrous cells.

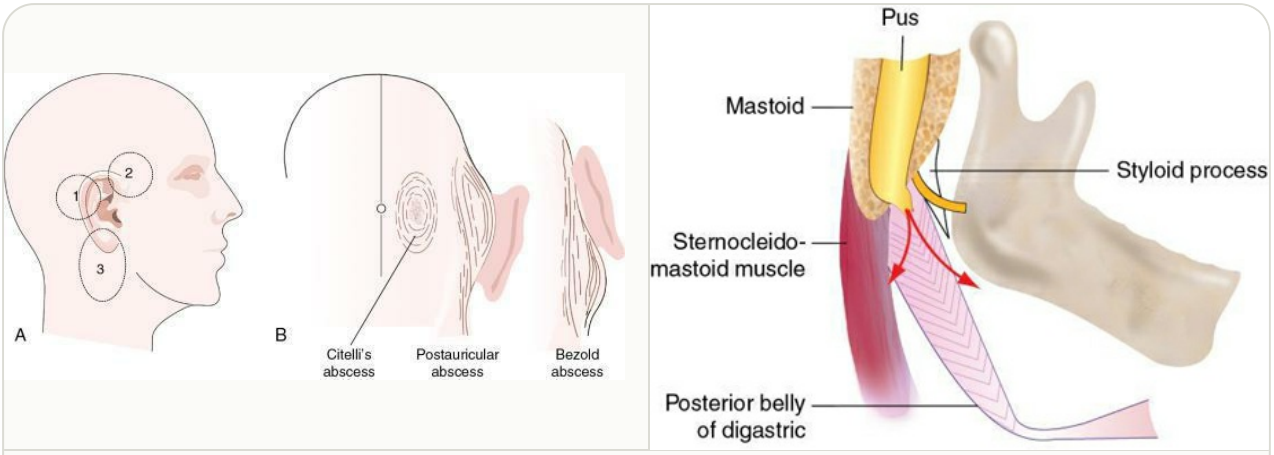


FIGURE 4 First, abscesses around the mastoid: postauricular, zygomatic and Bezold, with Citelli's abscess. Second, a Bezold abscess, where pus bursts through the medial side of the tip into the neck.

Dhingra, figs 12.4 and 12.5



FIGURE 5 A mastoid abscess that has burst through the skin.

Dhingra, fig. 12.3

5 CHAPTER FIVE Diagnosis

Mainly clinical, confirmed by imaging.

5.1 Blood tests

Full blood count with raised white cells, inflammatory markers, and blood cultures. Pus from the ear or myringotomy is sent for culture.

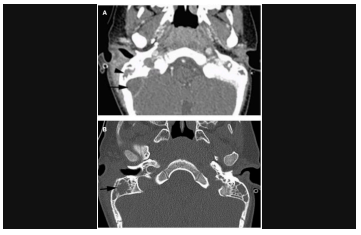
5.2 Imaging

5.2.1 CT temporal bone with contrast

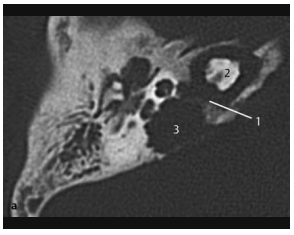
Shows coalescence (loss of the septa), cortical erosion and abscesses, and looks for intracranial spread. An opaque mastoid alone is not mastoiditis.

IMAGING IN THE APP 3 cases

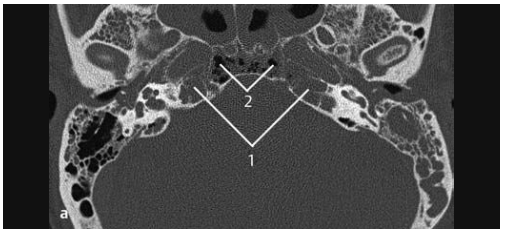
Otomastoiditis, petrositis and skull base infection on CT and MRI.



Otomastoiditis and its complications
Infection & inflammation · 9 images



Petrositis
Infection & inflammation · 3 images



Infection of the Skull Base and Gradenigo Syndrome
Infection & inflammation · 3 images

5.2.2 MRI and MR venography

When an intracranial complication or sinus thrombosis is suspected.

5.3 Differential diagnosis

SWELLING BEHIND THE EAR

	DISTINGUISHING POINTS
Furuncle of the canal	No preceding otitis media, painful tragus and pinna movement, normal drum, no mucoid discharge
Suppurating postauricular node	No ear history, superficial abscess, mastoid not tender
Otitis externa with nodes	Canal swollen and tender, drum normal
Infected sebaceous cyst	Superficial, punctum, ear normal

6 CHAPTER SIX Management

Admit, drain, and decide on surgery.

6.1 First steps

ON CALL: ACUTE MASTOIDITIS

1. Admit; take blood and ear cultures.
2. Start intravenous third-generation cephalosporin, with anti-pseudomonal cover in chronic ears.
3. Examine the facial nerve, balance, neck and neurology.
4. Request contrast CT of the temporal bones and brain; MRI and MRV if intracranial spread is possible.
5. Perform a wide myringotomy with a grommet, and send the pus for culture.
6. Call your senior for an abscess, coalescence, or no response at 24–48 hours.

6.2 Medical treatment

Intravenous antibiotics, changed according to culture, with oral step-down to complete three to six weeks in total. Uncomplicated acute mastoiditis without coalescence often settles with antibiotics and myringotomy alone.

6.3 Surgery

6.3.1 Indications for cortical mastoidectomy

- Subperiosteal abscess.
- Coalescence on CT.
- Sagging of the posterosuperior canal wall or a positive reservoir sign.
- No improvement after 48 hours of adequate treatment.
- Any complication, or cholesteatoma.

6.3.2 Small children

A small subperiosteal abscess may be aspirated or incised, but cortical mastoidectomy is generally advised. The infant mastoid is small and vascular, landmarks are poor, and the facial nerve exits superficially.

6.3.3 The operation

Cortical (Schwartze) mastoidectomy removes the diseased cells and opens the antrum to the middle ear, leaving the canal wall and ossicles intact.

OPERATIVE GUIDES IN THEATRE 3 operations

Mastoid Surgery
Cortical mastoidectomy for coalescence, abscess or failure

Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation
Myringotomy and tube

Intracranial Complications of Otitis Media
Intracranial complications

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Coalescence:** what it means and how it is shown on CT.
- **Signs:** forward-displaced pinna, sagging canal wall, reservoir sign.
- **The abscesses:** subperiosteal, zygomatic, Bezold, Luc and Citelli, and where each points.
- **Indications for cortical mastoidectomy.**
- **Masked mastoiditis:** antibiotics without cure.

7.2 Common pitfalls

COMMON PITFALLS

- Treating an opaque mastoid on CT as mastoiditis without clinical signs.
- Relying on suprameatal tenderness alone.
- Missing masked mastoiditis in a child whose otalgia never settled.
- Forgetting the superficial facial nerve when incising behind an infant's ear.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 12 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Otomastoiditis and its complications

IMAGING FINDINGS

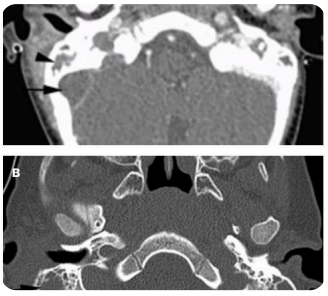
- Opacified middle ear and mastoid
- Coalescence: loss of the mastoid septa and cortical erosion
- Complications: subperiosteal abscess, sigmoid sinus thrombosis, epidural or brain abscess, labyrinthitis

TOP DIFFERENTIALS

- Uncomplicated effusion (septa intact)
- Cholesteatoma
- Langerhans cell histiocytosis or rhabdomyosarcoma in a child

CHECKLIST

- Contrast CT or MRI with venography when complications are suspected
- Check the sigmoid sinus, tegmen, labyrinth and petrous apex



Post-contrast CT shows (a) right coalescent mastoiditis (black arrowhead) and ipsilateral sigmoid sinus thrombosis (black arrow) – bot…



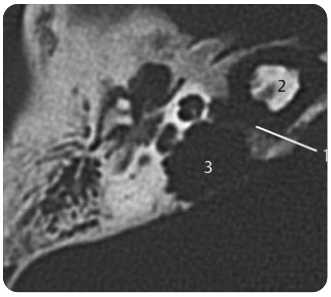
Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.

TEMPORAL BONE & SKULL BASE · INFECTION & INFLAMMATION

Petrositis

- Clinical symptoms may help differentiate between inflammatory disorders, asymptomatic pseudotumors, and other pathology. CT and MRI are complementary. Knowing whether the origin is intradural or extradural and the presence of bony changes help to establish the most likely diagnosis.

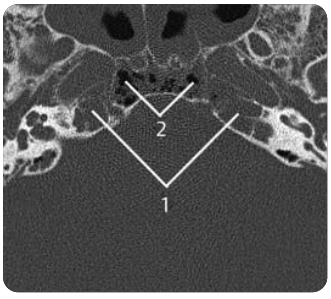


CT, axial. The internal auditory canal is seen (1), but is not clearly recognizable. Anteriorly, a rounded lucency in the petrous apex is visible…

TEMPORAL BONE & SKULL BASE · INFECTION & INFLAMMATION

Infection of the Skull Base and Gradenigo Syndrome

- In Gradenigo syndrome, the dural branches of the first division of the trigeminal nerve are involved in the symptoms of pain; the abducens nerve is paralyzed because of compression in the Dorello canal in the petrous apex. Pseudo-Gradenigo may be due to a nasopharyngeal carcinoma infiltrating the skull base.
- Malignant otitis externa may lead to skull-base infection. Immunocompromised and diabetic patients are prone to bacterial (e.g., Pseudomonas) or fungal (e.g.,) infections. Sequential radiographic and radionuclide (e. Aspergillus g., technetium and gallium) imaging may be used to monitor the efficacy of treatment. Complications may result from the spread of infection to adjacent intracranial or head and neck structures.



CT, axial. The petrous apex on both sides (1) is opacified. This could be because of normal fatty marrow but, considering the opacification in the…